

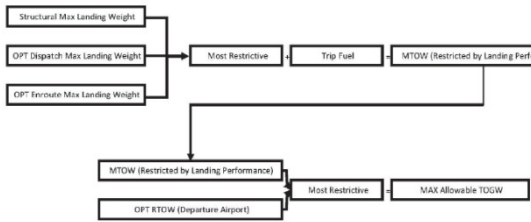
RWY CHG	
NOTAMS	Checked
OPT Take-Off Performance	Complete/Crosscheck
FMC/CDU Change:	Enter & Verify
• DEP/ARR – Rwy/SID/Route	
• LEGS – Review SID Wpt/Spd/Alt	
• EO SID	
• ACCEL HT & THR REDUCTION Ht	
• NAV RAD – As needed	
EMER & DEP Brief	Complete
TAKE-OFF Data: LOADSHEET	Enter & Verify
MCP:	Set
• V2, LNAV/VNAV/HDG/ALT	
TRIM	Set
FLAPS	As Needed
TAKE-OFF REVIEW	Complete
BEFORE TAKE-OFF CHX	Complete

Position	Minimum Flight Altitude
On airway	Minimum Terrain Clearance Altitude (MTCA)
Off airway	Minimum Grid Altitude (MGA)
Driftdown and/or Rapid Depressurisation Strategy (route or area specific)	As per strategy

DISPLACED THRESHOLD – GUIDANCE TABLE

This table shall be used when conducting an approach other than ILS to a temporary displaced threshold.

Displacement of Threshold	Delay to Descent Initiation Point – nm	Addition to All Crossing Altitudes in Feet (Except MDA)
< 300 m (984 ft)	No restrictions – use normal guidance.	
301-400 m (987-1313 ft)	0.2	60'
401-500 m (1314-1641 ft)	0.2	80'
501-600 m (1642-1969 ft)	0.3	100'
601-700 m (1970-2297 ft)	0.4	110'
701-800 m (2298-2625 ft)	0.4	130'
801-900 m (2626-2953 ft)	0.5	150'
901-1000 m (2954-3280 ft)	0.5	170'
> 1000 m (> 3280 ft)	See Note 1.	



	> 3.5° [- ISA+25°C]
	3.2° ⇒ 3.5° [- ISA +10 ⇒ +25°C]
	2.8° ⇒ 3.2° [- ISA -10 ⇒ +10°C]
	2.5° ⇒ 2.8° [- ISA -10 ⇒ -25°C]
	< 2.5° [- ISA -25°C]

PROHIBITED ITEMS
UN2809 (Mercury)
UN3090 (Metal) – Incl Grp 1A/B & PI 968
UN3480 (Ion) – Incl Grp 1A/B & PI 965

VISUAL REFERENCES:	RNP APPROACHES (FCOM SP4):	Min 180° 772 – 48m 773 – 56.5m
CAT IIIB: No Lights	* Verify FMC GP Angle not less than 0.01° of Charted	FUEL BURN. APU – 250kg/hr GE – 2t/Hr
CAT IIIB DH: 1 Light	* FAF ‘At or Above’ – Can be changed to ‘At’	
CAT IIIA: 3 Lights	* PM – RAD NAV INHIBIT – ON on REF NAV DATA Page	
CAT II: 3 Lights + Lat EL	* PM – Verify RNP on RNP PROGRESS Page	
CAT I: APPR/RWY	* PF/PM – TERR on 1 side	
AOTILS: 7 Lights	* PF/PM – Progress Page if No NPS	
	* PM or PF – Map display on 10NM (if no NPS)	

In-Flight Failed or Downgraded Aircraft Equipment

Navigation Application	Navigation Specification	Navigation Accuracy (NM)	Area of Application	eRM Designation	Required Equipment
RNAV	RNAV 10	10	Oceanic/Remote, Continental		2 FMC **/****
	RNAV 5	5	Continental, Arrival	“B-RNAV required”	1 FMC **/****
	RNAV 2	2	Continental, Arrival	“RNAV 2 required”	1 FMC **/****/****
	RNAV 1	1	Continental, Arrival, Initial/Intermediate Approach	“RNAV 1 required” or “P-RNAV required”	1 FMC **/****/****/****
RNP	RNP 4	4	Oceanic/Remote		2 FMC, 1 GPS
	RNP 2	2	Oceanic/Remote, Continental		1 FMC, 1 GPS
	RNP 1	1	Arrival, Initial/Intermediate Approach	“RNP 1 required”	1 FMC, 1 GPS
	RNP APCH	1 (0,3 in FAS)	Initial/Intermediate/Final Approach	“RNP Approach”	2 FMC, 2 GPS
	RNP AR APCH	1 – 0,1	Initial/Intermediate/Final Approach	“Authorization Required”	2 FMC, 2 GPS

Type		Minimum Cleared or Treated Runway Width (m)			
B777 (all)		30			
Classification	D	A1	A2	CA	EA
Scenario	Aircraft dire emergency				
	ETOPS/Drift-down and Depressurisation				
	Note 1				
	Medical Diversion			Note 2	
	Takeoff, Enroute and Destination alternate				

LVO Downgrades		
Airborne	LAND 2	Cat IIIA
	Auto Throttle	Cat IIIA
	SGL SRCE Display	Cat 1*
	Autopilot	Cat I*
	No Autoland	Cat I*
	* Cat II Manual Landing	
LMC	500kg Max	
ZFW	± 2t can req OFP	
FUEL	5% Calc +2000/-1000	

	NADP1	NADP2	ICAO A	ICAO B
EO ACCEL	1000 (OPT value if Higher)			
THR REDUCTION	1000'	1000'	1500'	1000'
ACCELERATION	3000'	1000'	3000'	1000'
TO-B	2LR/F/ER/ULR – upto 3000' 32°-53°			
APU-to-Pack	2LR/F/ER/ULR – up to 6900' PA			

FL	°C	FL	°C	FL	°C	FL	°C	DA = PA + (120 x [OAT – ISA])
0	15	110	-7	220	-29	330	-51	
10	13	120	-9	230	-31	340	-53	
20	11	130	-11	240	-33	350	-55	
30	9	140	-13	250	-35	360+	-56	
40	7	150	-15	260	-37			
50	5	160	-17	270	-39			
60	3	170	-19	280	-41			
70	1	180	-21	290	-43			
80	-1	190	-23	300	-45			
90	-3	200	-25	310	-47			
100	-5	210	-27	320	-49			

OCEANIC Clearance - WEST	
Shanwick	EGGX 90-30 min
Santa Maria	LPPO At least 40 min
Bodo	ENOB 90-30 min
Reykjavik	BIRD Fm Stavanger/Scottish – 25 min BIRD Fm Murmansk – 30 min BIRD Fm Edmonton – 45 min
Gander	From previous FIR 90-60 min
New York	KUSA From previous FIR
OCEANIC Clearance – EAST	
New York	KUSA
Gander	CZQX 90-60 min
Reykjavik	BIRD Fm Stavanger/Scottish – 25 min BIRD Fm Murmansk – 30 min BIRD Fm Edmonton – 45 min
Bodo	From previous FIR
Santa Maria	LPPO At least 40 min
Shanwick	From previous FIR

[] FLAPS DRIVE	< FL200 • FMC PREDICTIONS N/A (Perf) • LDG Dist/↓WT • ETOPS/RVSM
[] SLATS DRIVE	< FL200 • FMC PREDICTIONS N/A (Perf) • LDG Dist/↓WT • ETOPS/RVSM
[] FLAP/SLAT CONTROL	↑ EXT Time (15nm/5min) • Reduce to MLW • F20 LDG
[] FMC	(1) ↑ Appr & RTE Copy • LNAV can be re-engaged • No RNAV/RVSM • Vectors for Appr • ATC NAV Auth • (2) No VNAV / Man tune all Nav aids
[] FLIGHT CONTROLS	Keep Spd Window Open • No Autoland • AP Avail • F20 • X-Wind 20kt
[] STABILIZER	Keep Spd Window Open • No Autoland • AP Avail • F20
[] DOOR FWD CARGO	RVSM Affected • Poss CABIN ALT/STABILIZER/FIRE ENG/SVR DAMAGE • Oxygen Requirements?
[] ELEC AC BUS	APU Fuel Burn • Main Tank Fuel Jettison may be affected
[] CABIN ALTITUDE	PA/Lights/ATC • Aim for ALTN • 2000' to Level-off • LRC 300Kts • Step 0 • CRZ Alt
[] TIRE PRESS	T/O – Don't raise GR! (Eval 1 st) • Gr Synoptic • Advise ATC • Monitor HYD QTY
[] RESERVE BRAKES/STRG	Possible 'HYD PRESS SYS C' • Possible Altn Gr Ext • Do not Taxi
[] ENG FAIL	LANSA • ETOPS/RVSM • Fuel Balance
[] ENG OIL TEMP	Consider possible SE Scenario
[] FIRE CARGO	FWD Cargo: Equip Cooling Ovr • 6 Bottles (2 Immediate, 4 after 20min) • No bottles for other hold • Do Not open holds until all off
[] EQUIP COOLING OVRD	As soon as smoke a factor – Disregard Chx and continue to Smoke, Fire & Fumes

FTL's								
2 or More Acclimatised Crew					2 Crew Greater than 7 Hours			
Local Start Time	SECTORS				Scheduled Sector Time	ACCLIMATISED	UNACCLIMATISED	
	1	2	3	4				
0600 - 0759	13	12:15	11:30	10:45	Over 7 & ≤ 9 Hrs	2 Sectors	4 Sectors	
0800 – 1259	14	13:15	12:30	11:45	Over 9 & ≤ 11 Hrs	3 Sectors	4 Sectors	
1300 – 1759	13	12:15	11:30	10:45	Over 11 Hrs	4 Sectors	N/A	
1800 – 2159	12	11:15	10:30	09:45	DEPU – Max 8 DEPA – Max 1 (or family)			
2200 - 0559	11	10:15	09:30	09:00				
2 or More Unacclimatised Crew								
Preceding Rest Time	SECTORS				RFF (ER/LR)	DEP/DES	TO/ENr/DA	ETOPS/D&D
	1	2	3	4				
Upto 18hrs or Over 30hrs	13:00	12:15	11:30	10:45		9(8/7)	7	5
Between 18 – 30hrs	11:30	11:00	10:30	09:45	(777F)	6	6	4

Restrictions and Limitations				
		No medical certificate or letter required	Doctor/Midwife certificate or letter required	EK Medical Services acceptance required (MEDA)
Uncomplicated pregnancy	Up to end of 28 th week	✓	–	–
Uncomplicated pregnancy	Entering the 29 th week	–	✓	–
Single uncomplicated pregnancy	After the 36 th week	–	✓	✓
Multiple uncomplicated pregnancy	After the 32 nd week	–	✓	✓
Complicated pregnancy	–	–	✓	✓

Ground Speed	Rate of Descent
13___	650-700'/min
14___	700-750'/min
15___	750-800'/min
16___	800-850'/min
17___	850-900'/min
18___	900-950'/min

59-60 N1

Cat D – Code E

[Lido mPilot only]

The ATS route centerline is depicted in a different color according to the MTCA of the segment:

- Black: MTCA ≤ 10000ft
- Salmon: 10000ft < MTCA ≤ 14000ft **10% of PAX after 30min**
- Burgundy: 14000ft < MTCA **30% of PAX 14000' ≥ 15000'**



PERFORMANCE: 335t

30 000' - 280KIAS - 3° - 86.3 N1 - Cruise (Flaps Up)
20 000' - 270KIAS - 4° - 76.6 N1 - Cruise (Flaps Up)
5 000' - 263KIAS - 5.5° - 63.4 N1 - Holding (Flaps Up)
5 000' - 196KIAS - 5.5° - 75.5 N1 - Holding (F20, GR Dn, Vref30 + 20)
1 500 - 184KIAS - 1.0° - 70.5 N1 - Final 3° GS (F30, Gr Dn, Vref30 + 10)
Sea Level - 196KIAS - 14.7° - 3100'/min ROC (F20, Gr Up, G/A Thrust)

MARITIME DISTRESS:

14300
2182
12290
6215